

SALES ORBIS: AI-BASED SALES PIPELINE INTELLIGENCE AND DEAL WIN PREDICTION PLATFORM

PHASE I REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Over multi-stage projects, multiple suppliers, B2B opportunities, quotations, evaluations, activities, pipeline stages management and domain process data, B2B procurement and sales pipeline management is complicated and data intensiveness. Procurement team needs to understand supplier, commercial, evaluation, engagement, competition and project data to make reliable award decision. Existing procurement and customer relationship management platforms mainly provide descriptive information and opportunity prioritization, supplier evaluation and prediction results depend heavily on manual analysis, history knowledge and experience. This project, the Sales Orbis: AI-Enabled Sales Pipeline Intelligence and Deal Win Prediction Platform, is a full-stack ML/AI driven procurement and supplier data-to-Pipeline intelligence and decision support platform.

This ML platform adopts a dynamic pipeline snapshot-centric technique, that the information at a certain as of date can be used to predict a likelihood that the supplier will end up with the award, by eliminating information leakage from future stages and outcomes. This project features a total of 11 structured procurement and supplier data sources containing information about projects, suppliers, opportunities, quotations, evaluations, activities, stage history, project status and outcomes, as well as a leakage-safe Gold ML feature table with 34999 prediction snapshots for 24857 opportunities. The machine learning pipeline includes data validation, data pre-

processing, feature engineering, project-grouped data splitting, temporal validation, model training, probability calibration, threshold optimization, risk bands generation, model explainability and probability expected-value estimation. Six classification algorithms, Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, Cat Boost, are evaluated by ROC-AUC, PR-AUC, precision, recall, F1-score, log loss, Brier score and calibration error. The final chosen model based on calibrated Logistic Regression yields an award probability, forecasted predicted outcome, forecasted risk-bands, and risk-bands based probability-weighted expected procurement value for each opportunity snapshot, and provides a web flask app with authenticated role-based platform, pipeline dashboard, active and closed deals explorer, deal explainability, probability progression, forecast and report, data excel export, user management and audit. The system leverages historical procurement data to produce structured predictive insights, enables earlier pipeline risks detection and procurement actions improvement, and helps to optimize B2B deal priorities and vendor award decisions. This report details introduction, related work, system design, data-structure, feature engineering and leakage-protection, ML modelling, web application modules, model evaluation and discussion. It demonstrates the benefits of the implemented leakage-safe predictive model with the procurement data-to-rational pipeline intelligence platform, and

proposes future work, including live data connection, next-best-action decision recommendation and enhanced predictive analytics.

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TABLE OF CONTENTS

ABSTRACT	iii
LIST OF TABLES	ix
LIST OF FIGURES	xi
LIST OF ABBREVIATIONS	xiii
1. INTRODUCTION	1
1.1 GENERAL	1
1.2 OBJECTIVE	4
1.3 EXISTING SYSTEM	5
1.4 PROPOSED SYSTEM	7
2. LITERATURE SURVEY	10
2.1 LITERATURE REVIEW	10
2.2 COMPARISON AND DISCUSSION	13
2.2.1 RESEARCH GAP	15
2.2.2 PROPOSED CONTRIBUTION/NOVELTY	16
2.3 CONCLUSION	17
2.4 RESEARCH QUESTIONS	18
3. SYSTEM DESIGN	20
3.1 SYSTEM ARCHITECTURE	20
3.2 SYSTEM REQUIREMENTS	22
3.2.1 SOFTWARE REQUIREMENTS	22
3.2.2 HARDWARE REQUIREMENTS	25
3.3 DATABASE DESIGN	26

3.4 DEPLOYMENT ARCHITECTURE	29
4. PROJECT DESCRIPTION	31
4.1 METHODOLOGY	31
4.2 MODULES	32
4.2.1 PIPELINE DASHBOARD & ANALYTICS MODULE	32
4.2.2 ACTIVE DEAL INTELLIGENCE MODULE	32
4.2.3 DEAL WIN PREDICTION & RISK ASSESSMENT MODULE	33
4.2.4 SALES FORECASTING & REPORTING MODULE	34
4.2.5 USER MANAGEMENT & ACCESS CONTROL MODULE	34
4.2.6 AUDIT TRAIL & SYSTEM MONITORING MODULE	35
5. IMPLEMENTATION AND RESULT DISCUSSION	41
5.1 API ENDPOINT SUMMARY	41
5.2 ADMINISTRATIVE DATA MANAGEMENT	43
5.3 END-TO-END SYSTEM DEMONSTRATION	43
5.4 FUNCTIONAL VALIDATION	43
5.5 SYSTEM OUTPUTS AND DECISION SCENARIOS	44
5.6 IMPLEMENTATION DISCUSSION AND OBSERVED ISSUES	44
5.7 LIMITATIONS	45
5.8 SUMMARY OF CONTRIBUTIONS	45
6. CONCLUSION AND FUTURE WORK	47
6.1 CONCLUSION	47
6.2 FUTURE WORK	48
REFERENCES	51

LIST OF TABLES

TABLE NO.	TABLE NAME	PAGE NO.
3.1	Backend software stack and libraries	17
3.2	Frontend software stack and libraries	18
3.3	Hardware / deployment requirements	19
3.4	Database entities and their purpose	21
5.1	Admin: Manufacturers register	32
5.2	Admin: Seller / Transaction register	33
5.3	Ownership Transfers register	34
5.4	Admin API endpoints	34
5.5	Global Claims Ledger	44

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1	System architecture diagram	15
4.1	Authentication & Access Control Module	27
4.2	Dashboard & Business Intelligence Module	28
4.3	Deal Lifecycle Management Module	29
4.4	Sales Forecasting & Predictive Analytics Module	29
4.5	Reporting & Document Generation Module	30
4.6	User Administration Module	30
4.7	Audit & Activity Monitoring Module	31

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
B2B	Business-to-Business
CSV	Comma-Separated Values
DB	Database
EDA	Exploratory Data Analysis
JSON	JavaScript Object Notation
HTML	HyperText Markup Language
PR-AUC	Precision-Recall Area Under the Curve
ML	Machine Learning
RF	Random Forest
SHAP	SHapley Additive exPlanations
F1	F1 Score
SQL	Structured Query Language
UI	User Interface
UML	Unified Modeling Language
XGBoost	Extreme Gradient Boosting
CRM	Customer Relationship Management
LR	Logistic Regression
RFP	Request for Proposal

CHAPTER 1 INTRODUCTION

1.1 GENERAL

Managing multiple projects, suppliers, opportunities, quotations, evaluations, activities and pipeline stages requires a lot of effort before a supplier is selected for a specific requirement. During a typical procurement and sales pipeline, for each opportunity, procurement teams need access to project and supplier details, estimated budget, quoted value, evaluation scores, quotation details, activities, market information, current pipeline stage and status of a project. In the existing procurement and sales pipelines, much of this information is captured for tracking and reporting as well as manual analysis. It becomes hard to prioritise opportunities and assess supplier outcome as the number of projects and suppliers grows, especially when potential risks need to be identified before the actual outcome of procurement is known. Additional procurement historical data allows to transition from retrospective to predictive procurement analytics. Whereas an opportunity can be assessed only after its final result is known, a predictive system can leverage the knowledge that is present at some stage in the procurement lifecycle, to predict the likelihood that the supplier will win the award ultimately. A predictive procurement prediction system should be meticulous in differentiating between data that was known at the point of prediction and data that might have arrived after the point of prediction. This aspect is especially sensitive for procurement prediction as the final results, later stage transitions, subsequent steps and other state can result in data leakage and over estimates of model performance.

In this project, a probabilistic, snapshot-based prediction problem is created by assigning each opportunity to multiple points in its procurement process. This future

estimate of $P(\text{supplier eventually awarded} \mid \text{available as of date info})$ is estimated using only information available as of each snapshot_date. Leakage-safe Gold ML feature table has a total of 34,999 snapshots of 24,857 opportunities, 5,000 projects and 1,200 suppliers. The data set consists of snapshots within the period of 20 January 2025 to 15 January 2026. In the dataset, 15,285 (43%) of the 34,999 snapshots are labelled; and they represent 5,143 decided opportunities, with the remaining 19,714 rows (57%) for the open pipeline to be scored for prediction. The target variable for supervised learning, is_awarded, validates as correctly capturing the procurement outcome within each opportunity.

Sales Orbis: AI-Driven Sales Pipeline Intelligence and Deal Winning Propensity Module

This project has driven the following end-to-end predictive intelligence pipeline for B2B procurement and supplier intelligence. The combination of dataset validation, processing, leakage safe feature engineering, project-grouped and time-based data splitting, machine learning, probability calibration, risk assessment, explainability, and dynamic stage-wise analysis enables us to estimate the expected procurement value based on a single deal's predicted win likelihood, the current stage and the pipeline to-go. This feature engineering step also applied project, supplier, commercial, evaluation, pipeline, engagement, and competitor information – along with derived features such as project duration, quotation-to-budget level scores, activity rate, response rate, pitch/negotiation activity, stage progression rate, supplier composite score, and availability of quotations, evaluations, competitors and activities. The modelling pipeline compares Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM and Cat Boost models and evaluates model performance on the test dataset with ROC-AUC, PR-AUC, precision, recall, F1-score, logloss, Brier score and Expected Calibration Error (ECE). The final prediction pipeline in deployment is Logistic Regression with sigmoid calibration, using a classification threshold of 0.225. On the project-grouped test set, the sigmoid calibrated model achieves a

ROC-AUC of 0.8318, PR-AUC of 0.4154, precision of 0.3941, recall of 0.5574 and F1-score of 0.4617 at the chosen threshold. The calibrated probabilistic prediction scores are used to assign risk levels based on risk cut-points derived from the observed base award rate in the calibration set. Our prediction output is converted into actionable procurement insight with award probability, predicted award, risk and probability weighted future procurement value. The project also delivers with SHAP-based explanations for each prediction, which highlights the positive and negative features leading up to the specific prediction, and probability trajectory, which reveal the predicted award probability of an opportunity across the different stages of its available snapshots. Both of these components enable the system to display an output value that, in addition to the predicted value, presents insights into the defining features and stage of the procurement. Procurement pipeline access to machine learning outputs across multiple projects to realize strategic, market and supplier insights. An end-user role-aware web application integrating the outputs into one place to access the procurement pipeline dashboard, active-deal explorer, deal details, closed-deal analysis, sales forecast, reports, Excel/CSV downloads, user management and audit trail. Deal-level views displaying win probability, risk level, expected value, factors for prediction, probability Path, and project, supplier, commercial, engagement information. The web app uses the model outputs produced with the ML pipeline and is not retrained with the web interface during web app usage. Hence, the task of the project is to bring together procurement data engineering, leakage safe feature engineering, a machine learning award prediction model, probability calibration, risk, explainability, and pipeline intelligence in a platform for one decision support system. To be clear, the project does not propose any new machine-learning algorithms; it combines a standard predictive model, a procurement data pipeline, and an operational web application to turn historical procurement data into actionable predictive insights. The implemented system

supports opportunity ranking, leak prediction earlier in the process, and data-driven award decisions. Additional features such as real-time updates or next-best-action predictions could be added as defined in the project scope.

1.2 OBJECTIVE

The objective of this project is to design and implement a working, end-to-end AI-based sales pipeline intelligence and deal win prediction platform with the following concrete goals:

- Predict the probability of supplier award for procurement opportunities using the information available at each prediction snapshot, helping identify opportunities with higher or lower likelihood of successful contract award.
- Build a leakage-safe machine learning dataset by integrating project, supplier, opportunity, quotation, evaluation, activity, pipeline-stage, outcome, and project-status information while ensuring that future information is not used for prediction.
- Engineer meaningful procurement features from project, supplier, commercial, evaluation, pipeline, engagement, and competitive information to capture factors that influence supplier award outcomes.
- Develop and compare multiple machine learning models, including Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost, using appropriate classification and probability-quality metrics to identify a suitable predictive model.
- Generate calibrated award probabilities and risk levels so that predictions can be interpreted as decision-support information rather than only as binary

classifications. The system also estimates probability-weighted expected procurement value for open opportunities.

- Provide explainable prediction insights by identifying the factors contributing to individual predictions and presenting probability trajectories across available opportunity snapshots, allowing users to understand how an opportunity's predicted outcome changes over time.
- Provide a web-based procurement intelligence platform through which users can view the sales pipeline, active and closed deals, individual opportunity details, forecasts, prediction results, risk levels, reports, and downloadable data.
- Support data-driven opportunity prioritization and early risk identification by transforming historical procurement information into actionable predictive insights that can assist procurement and sales teams in evaluating supplier opportunities and potential award outcomes.

1.3 EXISTING SYSTEM

In many organisations, the management of the B2B procurement and sales pipeline is not necessarily controlled through a single predictive intelligence platform, but is managed across CRM, procurement, spreadsheet and reporting systems. Data relating to projects, suppliers, opportunities, quotes, assessments, activities, pipeline stages and the status of a project may be spread across a variety of records and operational processes. While systems like this are valuable for recording and monitoring procurement-related information, such approaches are limited to providing past and present descriptions of an opportunity's situation, and don't assist procurement teams in identifying which opportunities should be prioritised, and

whether a supplier is likely to receive a contract. In typical procurement systems, metrics in areas such as opportunity prioritisation and sales forecast are often predominantly a result of survey based judgement, old experience and occasional reporting than on forward looking probability calculation. In addition, information on supplier performance, evaluation, commercial terms, customer engagement, competition and pipeline progression, etc. is likely to be viewed independently. This makes it challenging to combine all the information to come up with a single, reliable prediction of the underlying potential sale, and therefore also in identifying at-risk opportunities. Other Limitations Another challenge, especially when using time-based procurement data, is the risk of making forecasts that are unreasonably accurate. A basic form of the model we use to forecast may inadvertently have access to information that was observed after the point of the forecast – like anything downstream of the current pipeline stage, for example, or activity, or ultimate outcome. The way that the descriptive workflows are currently set up don't lend themselves to provide a leakage safe, snapshot-style solution that can leverage only that information that was available at the time of the forecast. Conventional systems can also offer little explanation and probabilistic decision support for individual opportunities. Potentially, users could see the current stage, supplier information, quotations, evaluations or activities, but not a normalized award probability, risk level, economic value of the procurement, probability evolution or an explanation of the factors driving an individual prediction. Users would therefore have to manually integrate information from across the procurement process before making a judgement.

Another missing part in traditional CRM and procurement software is an integrated predictive workflow. Data wrangling, feature engineering, model evaluation, probability calibration, risk analysis, explainability and reporting of the pipeline are

often separated and not translated into a single end-to-end workflow. As a result, the current approach is very good for record keeping and descriptive reporting, but provides very little automation for scoring ongoing supplier awards, uncovering early risks and prioritizing deals on the basis of probability. Sales Orbis remedy this situation by bringing leakage-free procurement data wrangling, machine learning prediction, probability calibration, risk analysis, explainability and web-based pipeline intelligence into a single integrated platform.

1.4 PROPOSED SYSTEM

Sales Orbis's AI-powered procurement intelligence pipeline combines guaranteed procurement data and supplier data into a single pipeline, which delivers real-time supplier award prediction, risk modeling, forecasting, and decision support reports. All data elements related to project, supplier, opportunity, quote, evaluation, activity, pipeline stage history, project stage outcomes, and project stage status history are combined into a hierarchical machine learning pipeline. A leakage-free Gold ML feature table provides a snapshot representation of procurement opportunity as at date from all past snapshot dates, capturing all features as at date. Prior to training a model, this data preparation and feature engineering pipeline validates, preprocesses, constructs features, and controls leakage. Prediction data set is constructed by including project, supplier, commercial, evaluation, pipeline, engagement, and competition features. We also compute additional features in relation to quote and budget relationships, in evaluation, activity and response behaviours, in pipeline movement, in negotiation activity, and in tendering behaviour, and procurement information. Project-grouped, time-based validation is used to further reduce the risk of leakage between opportunities within the project and over time. Predictive modelling stage – After feature engineering, the project proceeds to a predictive modelling stage, where a number of different machine learning algorithms are tested,

including: Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and Cat Boost. The model is evaluated on several metrics, including ROC-AUC, PR-AUC, precision, recall, F1-score, log loss, Brier score, and calibration error. The final prediction system produces a probability of award for each opportunity snapshot based on calibrated Logistic Regression. The probability is then translated into a predicted outcome and risk using the chosen decision threshold, alongside the risk-band configuration. This calibration produces probability-weighted expected procurement value for additional analysis of expected open pipeline value. Moreover, the platform is also enabled with explainability and temporal capabilities for single opportunities. SHAP explanations reveal the features which shape the individual predictions while probability trajectories display how the probability of awarding the contract varies across the available snapshots of an opportunity. The evolution of each probability can now be observed in parallel to its corresponding risk, value, explanation and procurement stage. Web Application A web app is developed on top of the predictive pipeline, which allows access to the results through a single application. The application consists of a dashboard for analyzing the whole pipeline, an active-deal explorer, the details of a deal, closed-deal analysis, sales forecast, reports, excel/csv downloads, users management, and auditing. The individual opportunity view consolidates project, supplier, commercial, engagement, prediction, risk, expected-value, explanation, and probability-trajectory data, enabling analysis of a single opportunity from different angles. The first sections of this report contain related work in B2B sales prediction, sourcing data, procurement analytics, outcome prediction with machine learning, explainable AI, probability calibration, and predictive decision support (Chapter 2); the architecture, data flow, technology stack, and system design are then described (Chapter 3); followed by descriptions of the data engineering, feature engineering, prediction, risk analysis, forecast,

explainability, and application modules (Chapter 4); implementation, model evaluation, web app functionality, and discussion of results are next (Chapter 5); finishing with a discussion of the results, limitations, and future plans for the project (Chapter 6). This task documents the integration of the foremost data engineering on procurement data and prediction intelligence for the project, not the creation of a new ML algorithm. The application, as given, is a workflow integrating the leakage-safe construction of snapshot features, their feature engineering, side-by-side model comparison, probability calibration, risk score derivation, explainability, expected-value computation, and web-based inspection of the data pipeline. The solution is targeted toward earlier detection of procurement risks, prioritization of opportunities, and data-driven vendor awarding; future phases could deliver real-time updates and next-best-action decisions.

CHAPTER 2 LITERATURE SURVEY

This chapter surveys the previous work regarding B2B sales prediction, procurement outcome prediction, supplier selection, machine learning for sales prediction, explainable AI, and risk prediction. The survey covers the purpose of the work, database, system, method and model, results and limitation of the existing research. The survey is used to find out the research gap and bases of the design of the proposed predictive procurement intelligence system.

2.1 LITERATURE REVIEW

Some studies of this type already use predictive analytics machine learning, as well as the technology of artificial intelligence, for the purposes of developing the sales function as well as for creating sales channels. Yan et al. proposed a predictive sales pipeline analytics framework for B2B sales, predicting the probability of winning in sales opportunities over time based on the opportunity lead profile and seller activities [1]. Yan et al. also developed a regression-based framework for B2B sales pipeline win-probability prediction considering data sparseness and business noise as well as market volatility [2]. Those 2 studies utilized historical opportunity and activity information for win-probability prediction for sales pipeline management. The second line of research is attempting to forecast B2B sales and opportunities with machine learning. Eitle and Buxmann created ML models for sales pipeline management using real CRM data and the prediction of sales opportunities in different pipeline stages [3]. Lambert analyzed ML-based methods for calculating the B2B sales opportunity win propensity that include Random Forest, general linear models, boosting, decision trees, and neural network [4].

Rezazadeh introduced a generalized predictive modelling framework using Azure Machine Learning to estimate the probability of winning new B2B sales opportunity

and the corresponding decision boundary [5]. These approaches suggest that prediction can work as a complement to a CRM system to estimate the probability of opportunity success. Additional research has explored the subject of sales success prediction and lead conversion by the use of business data. Mortensen et al. explored the use of statistical modelling and machine learning to determine the factors associated with B2B sales success and to predict the win propensity of individual opportunities [6]. Woniakowski used machine learning and AutoML to B2B sales leads, and studied the relation between the predicted lead outcomes and actual business outcomes [7]. The study also explored the use of probability-weighted opportunity value to estimate sales potential.

The results of these studies are particularly applicable to systems that predict both probability of success and the business value of an opportunity, rather than simply the binary "win or lose" outcome. Importance of the topic The importance of sales forecasting and prediction in general have been raised in the broader context as well. Ahaggach et al. performed a systematic mapping study of sales forecasting methods, and highlighted main methodological advancements and trajectories of future research in the area [8]. Dominic et al. described the connection of management information systems and predictive analytics, and outlined how historical business data and machine learning can serve managerial decision making and sales prediction [9]. Kasem et al. examined customer profiling, segmentation, and sales prediction using artificial intelligence for marketing decisions, by integrating customer-level analysis with sales forecasting [10]. Opportunities scoring and intelligence in CRM Besides the investment in research for building an intelligent sales platform, there are specific studies concerning opportunity scoring and CRM intelligence: Nuthakki developed a new method for smart opportunity scoring (SOL-OMP) of sales opportunities in CRM sales cloud using opportunity history, sales

activities and engagement related features [11]. Yan et al. research on the predictive pipeline and Raja research on sales forecasting of CRM using Artificial Intelligence, also advocate for the benefits of pipeline intelligence in a CRM context [12]. These works underline that synergy between the two elements of predictive models and operational CRM functionality allows turning traditional CRM solutions for sales tracking into intelligent systems. More complex machine-learning modelling architectures for B2B sales forecasting and relationship modelling have been proposed. For example, Henna et al. developed a spectral graph convolutional network method for B2B sales forecasting and customer relationship management (CRM) that makes use of graph-based representations of business relationships, and compared the approach to more traditional machine-learning models [13]. This work shows that there is promise in more sophisticated architectures for capturing complex relationships within business data, but such methods will be more complex to implement than conventional machine-learning pipelines that learn from tabular data. Some of the studies not only focused on the modelling but looked into the actual sales process and execution of the pipeline. Sheahan constructed and empirically validated a Sales Pipeline Execution Process model for the early phase of the B2B sales process using SME and case company interviews [14]. It sheds light on the key role of systematic pipeline development and early stage sales activities, offering a process-oriented view in addition to machine-learning solutions, in showing the significance of systematic sales activities and stage management in the pipeline intelligence. So, the literature reviewed suggest that machine learning has been used to successfully predict B2B opportunity wins, lead forecasting, lead scoring, CRM analytics, pipeline management and business decision support. Yet the majority of the developed methods look at one capability, or a limited set of capabilities. The literature reviewed do not clearly bring together leakage-safe dynamic snapshot prediction, feature engineering of procurement and suppliers,

calibrated award probability, risk analysis, explainability, probability based expected procurement value, and web based pipeline intelligence all in one platform. Thus, the focus of Sales Orbis is on building an end-to-end AI-enabled sales pipeline and procurement intelligence platform. The ensemble will include historical procurement & supplier data, time-sensitive prediction snapshots, machine-learning award prediction, probability calibration, risk-calibrated award prediction, explainability, demand forecasting, and web-based decision support for procurement and sales pipeline management. This end-to-end product is aimed at taking research activities from predictive models toward a comprehensive, integrated solution.

2.2 COMPARISON AND DISCUSSION

When analysed collectively, the three emerging themes in the literature review have had a significant impact on the Sales Orbis system development. 1. Research on B2B sales in the literature suggests that data about opportunities, pipeline stage, activities, and characteristics of businesses can be utilised to make win probability estimates for a sales or deal [1] - [5] (e.g. Yan et al., Eitle and Buxmann, Lambert, and Rezazadeh's B2B opportunity prediction and sales pipeline management methods). Based on this Sales Orbis utilizes a snapshot-based approach where prediction is made at the time of each opportunity snapshot, allowing to infer a supplier's chance of being selected for a procurement contract by modelling the predictive attributes available as at each prediction date, rather than focusing just on the actual outcome [6]. Second, the fact that the literature provides is that the ability of predictive analytics is boosted if the model results are transferred into business-oriented information rather than just simply classification results. The prediction results of B2B sales success [6], lead prediction [7], opportunity scoring [8], and sales forecasting [9]-[12] are mainly all about win probabilities, opportunity value, scoring, and forecastings which could help users make a decision. They are realized

in Sales Orbis by generating award probability, prediction status, risk level, probability-based expected procurement value, and prediction explanation. Thus, in this project, machine learning will be used not only to classify procurement opportunities but also to provide information that helps users to review and prioritize procurement opportunities. Third, the consensus studies illustrate how predictive modelling needs to be embedded in well-defined business processes and operational systems. Work on sales pipeline execution, CRM analytics and advanced B2B forecasting demonstrates how prediction is most valuable when integrated with the pipeline and decision setting [11]-[14]. This was instrumental in designing Sales Orbis as a fully integrated web-enabled platform, not just a standalone machine-learning model. The system integrates procurement data preparation, feature engineering, prediction, forecasting, risk assessment, explainability, reporting and user interaction in a single pipeline. The major variations of the reviewed systems from the Sales Orbis are in the scope of integrating and the way the systems are (mainly) implementation for sales-oriented tasks. The B2B sales prediction research is mainly targeted at opportunity win prediction, lead-to-sale conversion, sales forecasting or pipeline scoring [1]-[7]. The CRM-focused research projects are mainly based on integrating the predictive analytics into sales processes and customer relationship management processes [11], [12]. Other projects utilize the advanced modelling methods like graph-based neural networks to represent complex business relation structure [13]. Process-oriented researches are mainly concerned on sales pipeline executing and managing practices rather than adopting the complete prediction platform [14]. These research projects provide important pieces of the actual platform, but, on the whole, they do not implement the complete workflow. What sales orbis is bringing to the table is not a revolutionary new machine-learning algorithm. Instead, it is bringing a range of existing algorithms together and applying them within a standardised decision-making platform, the

intelligent pipeline intelligence platform. This platform integrates multi-source procurement and supplier data, leakage-safe dynamic snapshots, feature engineering, various machine-learning models, probability calibration, risk assessment, explainability, estimated procurement value, forecast, and web-based reporting. The common decision-support environment is where the predictive outcomes can be explored at opportunity level and used to drive systematic pipeline interrogation.

2.2.1 RESEARCH GAP

Our literature search in section 2.1 has revealed that the topics of B2B sales forecasting, opportunity scoring, sales prediction, CRM analytics and ML based decision support have been the subject of much research; however, most of it focuses in a way which is constrained to isolated prediction or analysis, or to very narrow applications. B2B sales pipeline research including Yan et al. [1], [2], Eitle and Buxmann [3], Lambert [4], and Rezazadeh [5] has been predominantly targeted at opportunity win propensity and sales result prediction. Lead prediction and sales success studies [6], [7] explore the potential of historical business information for conversion and sales prediction. CRM and opportunity-scoring research [11], [12] provides solid evidence of predictive analytics applications for opportunity management and sales forecasting, while graph- or network-based models [13] demonstrate further capability for complex models of business relations.

Thus far, the literature review points to only a few closely related works that encapsulate all procurement-specific knowledge, combine the acquisition risk level and recommendation explanation, and use outcome and pipeline-stage information into a single acquisition-oriented dynamic prediction application. Most of the existing research emphasizes sales opportunities or accounts as opposed to customer

and project relationships, more so the unification of project, supplier, quotation, evaluation, activity, pipeline-stage, outcome, and project status information in one prediction system at the procurement pipeline level. Additionally, prediction most often is the output or the score, and the combining of the prediction with probability calibration, risk classification, expected acquisition value, and opportunity level explanations is less noted. This is the critical gap, then, and the one at the core of this work: it is this lack of a single workflow that performs multi-source procurement and supplier data integration, leakage-safe dynamic prediction point construction, business-relevant feature engineering, supplier award probability prediction, probability calibration, opportunity risk assessment, expected procurement value estimation, local explanation and global explanations of individual predictions, and reports these as part of a living, operational web-based pipeline intelligence platform, which this thesis aims to build. This thesis intends to fill the gap of Sales Orbis by integrating data engineering, leakage controlled feature construction, machine-learning based award prediction, probability calibration, risk assessment, explainability, prediction, reporting, and web-based pipeline management as one end-to-end system (Chapters 3,4).

2.2.2 PROPOSED CONTRIBUTION / NOVELTY

The innovation with Sales Orbis isn't some groundbreaking machine learning algorithm — rather, it's the integration of several existing machine learning and predictive analytics techniques into a singular AI-enabled sales pipeline and procurement intelligence system. The system We describe here is a new kind of dynamic, snapshot-based prediction workflow, in which information on past procurement activity and suppliers obtained as of a prediction date is used to predict the chance of a supplier being awarded a contract while accounting for information leakage from the future. Overview of the pipeline, which combines multi-source

procurement data, leakage safe feature engineering, machine-learning-based award prediction, probability calibration and risk estimation, expected procurement value estimation, and explainable prediction in one pipeline. Different classification models have been applied to choose the most relevant approach and subsequently generate award probability as well as prediction outcome. We then present how to translate the prediction probability into an interpretable risk and generate explanations on feature level to explain probability trend for opportunity-level analysis. Our additional innovation is connecting the pipeline prediction system with an online procurement intelligence platform, unlike the approach of keeping the machine-learning experiment as a separate stand-alone system. This platform offers views of the pipeline and opportunities, forecasts and predictive model details, reports, and data-enabled decision support. So, one can get from procurement raw information to predictive insights in one application. Thus, the main innovation of Sales Orbis is an integrated work-flow of leakage-controlled dynamic prediction, procurement and supplier intelligence, operations-ready calibrated award probabilities, risk-limited prediction and evaluation, explainability, expected-value calculation, forecasting, and web-based pipeline control, the latter two based on the work in [2] and [7] respectively. This work integrates the precursors we have given in previous papers into a single operating system.

2.3 CONCLUSION

Summary of Relevant Literature The literature presented in this chapter justifies the three broad design decisions inherent in Sales Orbis. The literature has shown the ability of machine-learning-based tools for B2B opportunity win prediction, sales forecasting, lead scoring, and pipeline management. This literature justifies, along with other chapters, the use of a forward-looking snapshot-based prediction approach, which considers the information available at a certain stage to estimate its

future supplier award. The literature supports use of multiple machine learning models, probability prediction, opportunity scoring, and business-oriented forecasting vs. descriptive pipeline data. As can be seen from the literature review for this work, the literature on B2B sales prediction, CRM analytics, forecasting and opportunity scoring, and advanced predictive modelling are tackled as individual research areas. The works reviewed by the literature review stages do not have the same exact combination of leakage-safe dynamic prediction, procurement and supplier feature engineering, probability calibration, risk calibration, explainability, expected procurement value, and web-based pipeline intelligence in a single operational platform. This is the problem that the sales Orbis solution attempts to fill. The implementation presented and the modules, methodology, machine-learning models, and results can be found from Chapter 3 onwards.

2.4 RESEARCH QUESTIONS

The research gap and proposed contribution outlined in Sections 2.2.1 and 2.2.2 have been converted into four research questions, to which the rest of this report is dedicated.

- RQ1: Is it possible for a leakage-proof, dynamic snapshot-based ML workflow, based on the data available at the time of prediction, to forecast the likelihood of supplier award in B2B procurement opportunities?
- RQ2: Can feature engineering develop multi-source procurement and supplierspecific information into informative predictive features, which help identify factors related to the eventual award outcome?

- RQ3: Would calibrated award probabilities, risk, explainable prediction insights and expected value of procurement be more useful decision-support information than a yes/no award prediction?
- RQ4: Can we embed predictive modelling, forecasting, risk assessment, reporting and opportunity-level analysis into a web-based platform that offers a complete and comprehensive environment for all AI-enabled sales pipeline and procurement intelligence?

RQ1-RQ4 are related to predictive modelling, feature engineering, decision-support and system integration with Sales Orbis. These questions are covered by the model design and implementation of the new system as explained in Chapter 3 and 4, and the experimental results and system demonstration in Chapter 5. Model performance, prediction behaviour, calibration, risk assessment, forecasting results and functionalities are explained by the results achieved from the implemented system. Known limitations and further areas for improvement are explained in Chapter 6.

CHAPTER 3 SYSTEM DESIGN

This chapter describes the architecture and design of the whole Sales Orbis platform including the system components, data flow, database, and machine-learning pipeline. The software and hardware environment for implementing the platform are explained. How the modules are integrated on the web-based procurement intelligence platform is also covered.

3.1 SYSTEM ARCHITECTURE

The Planned Procurement Pipeline Intelligence is a modular machine-learning architecture comprising: a data layer, feature engineering and preprocessing layer, model training and evaluation layer, prediction and business intelligence layer. The architecture has been optimised for the task of dynamic procurement award prediction, with leakage safe feature engineering pipeline and multiple machine-learning models. The `goldmlfeature_table.csv` is the main machine-learning dataset, with the raw procurement, supplier, quotation, evaluation, activity, stage-history, outcome, project-status data set being the support data used in features construction and validation. At the top level of the hierarchy of the prediction lifecycle, there are five stages. Load and validate the Gold feature table and data dictionary. Evaluate target definition, snapshot structure, missing values, label consistency, etc. Compose leakage-proof features based on information available up to each `asofdate`, including pipeline timing, commercial, engagement, supplier, project-status, availability, and competition features. Preprocess according to the models' specifications and train the model suite: Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and Cat Boost models using project-grouped splits and cross validation. Evaluate the models' performance on the validation data set using ROC-AUC, PR-AUC, calibrated log-loss, Brier score, and other

classification metrics; calibrate the probabilities; and select the threshold on the validation set. Use the selected model to predict award probabilities on procurement snapshots; convert them into predicted success/failure outcomes, four risk bands, and probability-weighted expected procurement values; and write prediction data sets to the archive to support probabilistic analysis, SHAP explainer plots, and the business intelligence report.

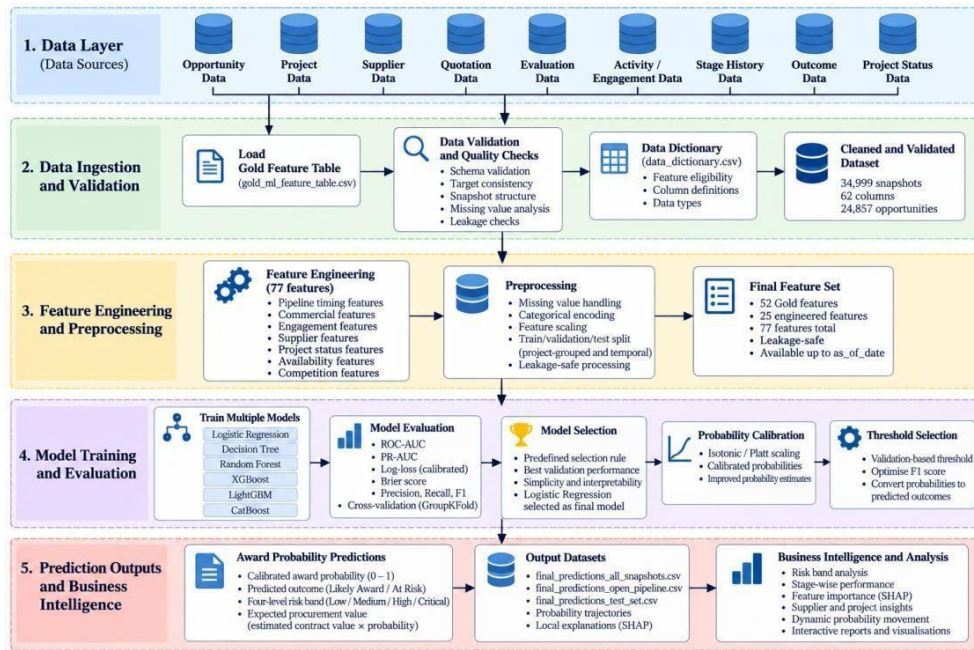


Figure 3.1. System architecture diagram

The above diagram illustrates at a glance, Opportunities for sales and procurement-related data sources in the top row, Data ingestion and validation layer with Gold Feature Table, and Data Dictionary, Leakage-safe and market-sensitive Feature Engineering and preprocessing layer with 77 features, Model training and evaluation layer with Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and Cat Boost, and The prediction and business intelligence layer producing the ultimately valuable deal-win intelligence outputs. Opportunity, project, supplier, quotation, evaluation, activity & engagement, stage-history,

outcome and project-status records are correlated and blended to make relatively more accurate and timely deal projections. The architecture depicts how the validated GOLD dataset is fed through the leakage-safe feature engineering and model-specific pipeline transformations to be separated into project-grouped and temporal evaluation sets. After training the models, they are evaluated using ROC-AUC, PR-AUC, calibrated log-loss, Brier score, precision, recall and F1-score and a pre-decided selection process chooses Logistic Regression as the best performing model, and consequently, the final model used to predict deal success probability. These predicted probabilities are calibrated and passed through the validation-based threshold to determine deal outcome labels and risk-bands (Low/Medium/High/Critical). The last step generates the essential deal intelligence using probability-based deal-win probabilities and outcome labels, Low/Medium/High/Critical risk-bands, probability-weighted expected procurement value, probability trajectories, shap explanations, stagewise model performance analysis, and business intelligence pack. The arrows in the figure indicate how information flows from raw sales/ procurement data to deal validation, feature engineering, model training, prediction and finally, actionable deal insights in the complete PL/SALES ORBIS platform.

3.2 SYSTEM REQUIREMENTS

3.2.1 SOFTWARE REQUIREMENTS

The backend has been designed as a machine-learning pipeline in Python. Table 3.1 highlights the most important libraries used within the backend for data processing, feature engineering, model building, scoring and explainability.

Component	Role
Python 3	Primary programming language used for the machine-learning pipeline and backend processing
Pandas	Loading, cleaning, transforming and analysing sales, supplier, opportunity and prediction datasets
NumPy	Numerical computation and array-based data processing
Scikit-learn	Preprocessing, pipelines, Logistic Regression, Decision Tree, Random Forest, evaluation metrics and cross-validation
XGBoost	Gradient-boosting model used for deal-win prediction
LightGBM	Gradient-boosting model used for advanced model comparison
CatBoost	Gradient-boosting model used for advanced deal-win prediction
SHAP	Model explainability, global feature importance and local prediction explanations
Matplotlib	Generation of model evaluation and analytical visualizations
Seaborn	Exploratory data analysis and statistical visualization

Joblib / Pickle	Saving and loading trained models and machine-learning pipeline artifacts
JSON	Storing configuration, validation results and metadata
Python-dateutil	Date and time processing for snapshot-based feature engineering

Table 3.1. Backend software stack and libraries

The frontend provides the presentation to the user for deal intelligence, prediction output and the analysis report. The project of web-based business intelligence front-end is built to provide an online business intelligence platform of the processed prediction information and sales pipeline analysis report. Details of the front-end presentation tools are listed in Table 3.2.

Component	Role
HTML5	Defines the structure of the web-based Sales Orbis interface and business report
CSS3	Provides styling, layout, cards, tables and visual presentation
JavaScript	Handles interactive behaviour, dynamic data presentation and user interactions
Chart.js / JavaScript Charts	Displays prediction metrics, risk distribution and analytical visualizations

HTML Tables	Presents deal-level prediction and pipeline information
Web Browser	Provides the execution environment for accessing the Sales Orbis interface and generated business intelligence report

Table 3.2. Frontend software stack and libraries

3.2.2 HARDWARE REQUIREMENTS

The required compute for SALES ORBIS is moderate because the system involves data pre-processing, feature engineering, training of machine-learning models, explainability via SHAP, production of predictions, and creation of HTML business-report pages. The project-based approach employs data-processing and machine-learning Python libraries, with tree-based models set to take advantage of multiple CPU threads where possible. No GPU is needed for the implemented pipeline.

Component	Recommendation
CPU	4 or more cores recommended for model training, data processing, and parallel execution of tree-based models.
RAM	8 GB minimum; 16 GB recommended for comfortable processing of the Gold dataset, feature engineering, model training, and SHAP analysis.

Storage	5 GB or more of free space for datasets, Python dependencies, trained models, generated predictions, reports, and analysis outputs.
GPU	Not required. The implemented models are configured for CPU-based execution; XGBoost, LightGBM, Random Forest and CatBoost use CPU threads.
Network	Internet connectivity is not required for the core ML pipeline once the Python dependencies and project files are available.
Development Environment	A standard Windows/Linux/macOS computer capable of running Python and the required machine-learning libraries.

Table 3.3. Hardware / deployment requirements

3.3 DATABASE DESIGN

SALES ORBIS is based on a procurement and sales-pipeline data backbone constructed from relational projects, suppliers, opportunity, quotation, evaluation, activity, stage history, outcome, and project-status-history datasets. These datasets embody the full opportunity lifecycle with the supplier from creation through bidding / quotation, evaluation, negotiation, approval, and finally award (outcome). The data has been designed to flow into the goldmlfeature_ table which is the primary machine learning data table containing time-based snapshots of each opportunity for analysis and machine learning. There are 34,999 snapshots for 24,857 opportunities across 5,000 projects and 1,200 suppliers in the Gold table.

Table	Purpose
procurement_projects	Stores one record for each procurement project, including project-level information such as category, department, type, budget, duration, priority, and status.
suppliers	Stores supplier information and supplier-level attributes such as category, region, rating, business history, risk, and performance-related information.
procurement_opportunities	Represents supplier participation in procurement projects and contains the main opportunity-level information used for deal/award prediction.
procurement_activities	Records activities and interactions associated with opportunities, including activity counts, responses, sentiment, and response-time information used to measure supplier engagement.
procurement_stage_history	Maintains the progression of an opportunity through procurement stages. It is also used to calculate leakage-safe stage features based only on information available up to the snapshot date.
supplier_quotations	Stores supplier quotation information such as quoted amount, delivery terms, payment terms, warranty information, and other commercial details.

supplier_evaluations	Stores supplier evaluation and assessment information, including technical, commercial, quality, delivery, compliance, and overall evaluation scores.
procurement_outcomes	Records the final outcome of procurement opportunities and provides the basis for the `is_awarded` prediction target.
project_status_history	Records changes in the status of procurement projects and supports project-status-related features used by the prediction pipeline.
gold_ml_feature_table	Consolidated analytical table containing time-based opportunity snapshots and the features used for machine-learning prediction. It contains 34,999 snapshots and 62 original columns.
model_dataset	Processed machine-learning dataset created after leakage-safe feature engineering, preprocessing preparation, and train/validation/test splitting.
prediction outputs	Stores generated prediction results such as award probability, predicted outcome, risk level, and expected procurement value for scored opportunities.

Table 3.4. Database entities and their purpose

The procurement data is linked primarily by projectid, supplierid and opportunityid. Activities, quotations, evaluations, stage history, and results are linked to their respective opportunities. The goldmlfeaturetable is the main ML dataset, with each row representing an opportunity snapshot at a given asofdate. The prediction target in the goldmlfeaturetable is the isawarded column, and open opportunities are only used for inference. Leakage-safe features have been created from stage progression, pipeline timing, commercial data, supplier history, activity engagement, and competition using only information that was available at each asofdate. This connects the raw procurement data, Gold feature layer, and ML prediction outputs.

3.4 DEPLOYMENT ARCHITECTURE

The SALES ORBIS pipeline is implemented using a Python machine learning and business intelligence pipeline. The sales and procurement pipeline data goes through the feature engineering, model training, evaluation, explainability and prediction pipeline. The trained models and prediction results are saved as output files using Python based serialization and CSV formats. The business intelligence output results are included in a standalone HTML report generated using the reporting module.

- ML Pipeline: Scripts in Python to process data, do feature engineering, train and evaluate the models, compute SHAP explainability of the models and generate the final predictions.
- Model Output: The models trained and its prediction results are stored in output directory of a project for reporting and further assessment.
- Deal Intelligence ReportA complete HTML report, dealintelligencereport.html, offers interactive data on active deals, closed deals, deal data, risk, forecast awards and sales forecasts.

- **Development Environment:** This program can be run on a local machine using Python installed with the necessary machine learning and data-processing libraries.

CHAPTER 4 PROJECT DESCRIPTION

Sales Orbis Use AI to Deal Win Prediction and optimization system that converts your past procurement data and supplier performance data into actionable business intelligence. Sales Orbis integrates data validation, leakage-free feature engineering, award prediction through ML, Probability calibration, Risk estimation, Explainable AI, Forecasting, and Business Intelligence reports into a single workflow on a platform. Using available information from each opportunity snapshot, that is, current pipeline information, early supplier award probability, risk classification, what is the expected procurement value, and reasoning explanation, ranked opportunity classification, prioritization and procurement planning becomes easy by using a single web-based interactive Intelligence platform.

4.1 METHODOLOGY

Sales Orbis' leverage a comprehensive, start-to-finish procurement intelligence pipeline, comprising data collection, quality assurance, feature engineering, predictive modelling, probability calibration, risk assessment, explainability, and business intelligence dashboards. Procurement project, supplier, opportunity, quote, evaluation, activity, stage history, outcome, and project status data points are extracted and merged into a single Gold ML Feature Table before leakage-resistant features are derived exclusively using project data available on the datetime of each prediction snapshot. A library of machine learning models are then trained using project-grouped validation, temporal validation, evaluated, and the most performant model select and calibrated to generate award probability. The predicted results are then turned into risk bands, predicted expected quantities to be purchased, probability curves, and explainable predictions that are then visualized through web based interactive tool we call ProcureIQ for decision support.

4.2 MODULES

4.2.1 PIPELINE DASHBOARD & ANALYTICS MODULE

The Pipeline Dashboard & Analytics Module is the main dashboard component of the Sales Orbis platform that enables monitoring of the procurement opportunities, open deals, vendor performance, deal progression, risk spread, and forecasted procurement value throughout the pipeline. The module displays the output of prediction generated from the deployed machine learning model and provides key performance indicators including open opportunities, forecasted procurement value, average win probability, risk-band spread, and statistics on opportunities per deal stage. Interactive visualizations, charts, summary cards, and analytics tables help check health of the opportunities in the pipeline, detect high-value opportunities, and risks early on. The integration of procurement data, machine learning predictions, and business intelligence visualizations helps procurement teams make data-driven decisions and prioritize deals based on the probability of award and forecasted business value.

4.2.2 ACTIVE DEAL INTELLIGENCE MODULE

The Active Deal Intelligence Module Once a procurement is underway, this module allows you to track the deal and view its projected outcomes over the course of the procurement process. It provides full details about all active deals including project details, supplier details, phase of procurement, award, and more. Opportunity Intell The module provides the probability, risk level, the expected dollar value of the opportunity, and a set of supporting predictive insights. Opportunities identified through the model calibration of the Logistic Regression are fed into the procurement data.

4.2.3 DEAL WIN PREDICTION & RISK ASSESSMENT MODULE

The Deal Win Prediction & Risk Assessment Module is the core intelligence module of the platform. Using the calibrated ML model, this module predicts the probability of supplier award for each procurement opportunity based on the data from the chosen prediction snapshot. The probability scores are then translated into prediction label (predicted award outcome) and assigned to one of the predefined risk categories (Low, Medium, High, Critical Risk).

Apart from the award prediction, this module also generates probability-weighted expected procurement values that can be used for opportunity ranking and forecasting purposes.

By employing the explainable AI techniques, this module also provides insights to identify the most important features contributing towards each prediction as well as probability trajectory analysis showing the changes in the probability of success as the opportunity progresses through the procurement lifecycle. These features enable procurement teams to identify upcoming at-risk opportunities, analyze the potential of procurement opportunities and make data-driven procurement decisions.

4.2.4 SALES FORECASTING & REPORTING MODULE

The Sales Forecasting & Reporting module enables users to analyze the procurement pipeline with a host of detailed reports that help supply best practice-based procurement analysis and strategic decision-making. The module consolidates a prediction table with opportunity & procurement data, opportunity risk profile, and the estimated procurements value to provide a set of pipeline forecast reports, business intelligence reports. Users can view the forecast pipeline stage data, pipeline revenue estimates, risk by opportunity, and historical opportunity performance on engaging dashboards, and report downloads. The module can generate daily pipeline reports, active and closed deal summary, procurement intelligence reports that predict future award potential and pipeline health, and make it easier to convert predictive analytics into procurement reports for management review by procurement teams and admin.

4.2.5 USER MANAGEMENT & ACCESS CONTROL MODULE

The Sales Orbis User Management & Access Control Module Our Sales Orbis User Management & Access Control Module is designed to ensure secure access to the platform and control user privileges according to roles within the organization. This module enables the administrator to create, activate, deactivate, and update users' accounts to access the system and grant access to specific features through role-based authorization system. The appropriate accesses are granted to different user roles to dashboards, deal intelligence, forecasting reports, Admin, and Audit Portal access. Authentication to safeguard the facility and safeguard against access to valuable procurement data. This module.

Provides the foundation for the safe functioning of the platform, while facilitating transparency and effective stakeholder collaboration in procurement.

4.2.6 AUDIT TRAIL & SYSTEM MONITORING MODULE

The Audit Trail & System Monitoring Module This Audit Trail & System Monitoring Module offers transparency & accountability & operational monitoring within the Sales Orbis platform. This module enables tracking and monitoring of all essential activities and transactions that take place within the user application, including login activity, user activity, report generation, data updates, changes, user activities, as well as system-level monitoring of user activities. All logs are stored with time, user or admin identity, description of activity, with information about affected or changed system modules. The module would support the system monitoring and administration by generating system monitoring logs of system status, user status, administrator activities, user activities, and activity trend analysis.

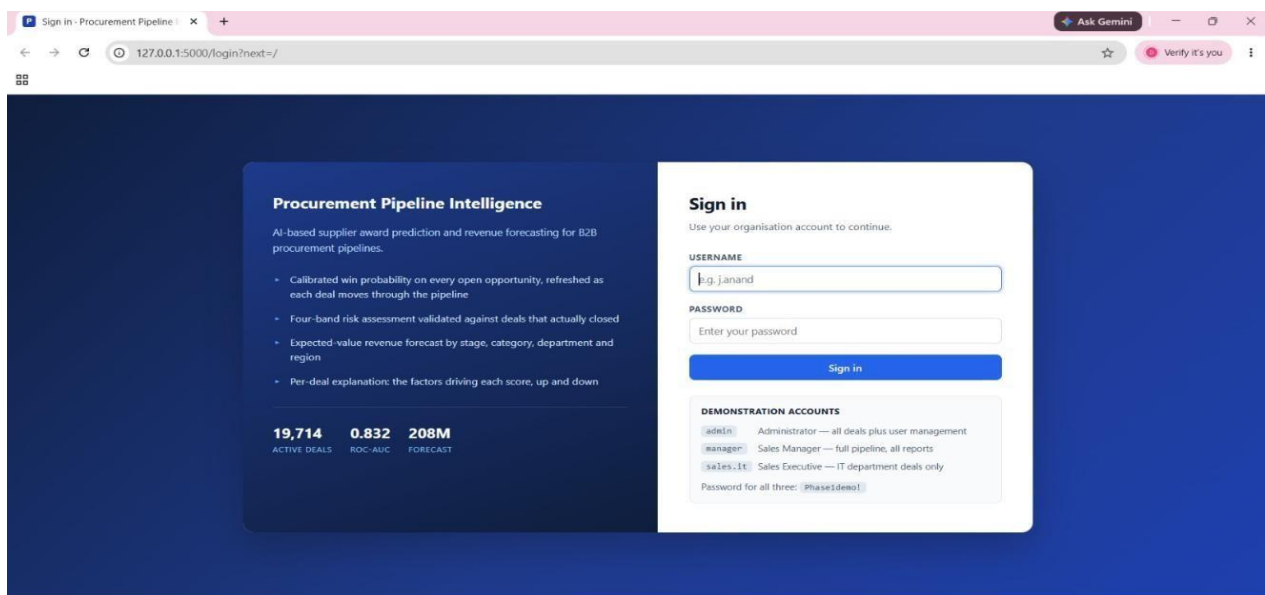


Figure 4.1. Authentication & Access Control Module

Provides secure user login and role-based access control. Ensures only authorized users can access procurement intelligence features.

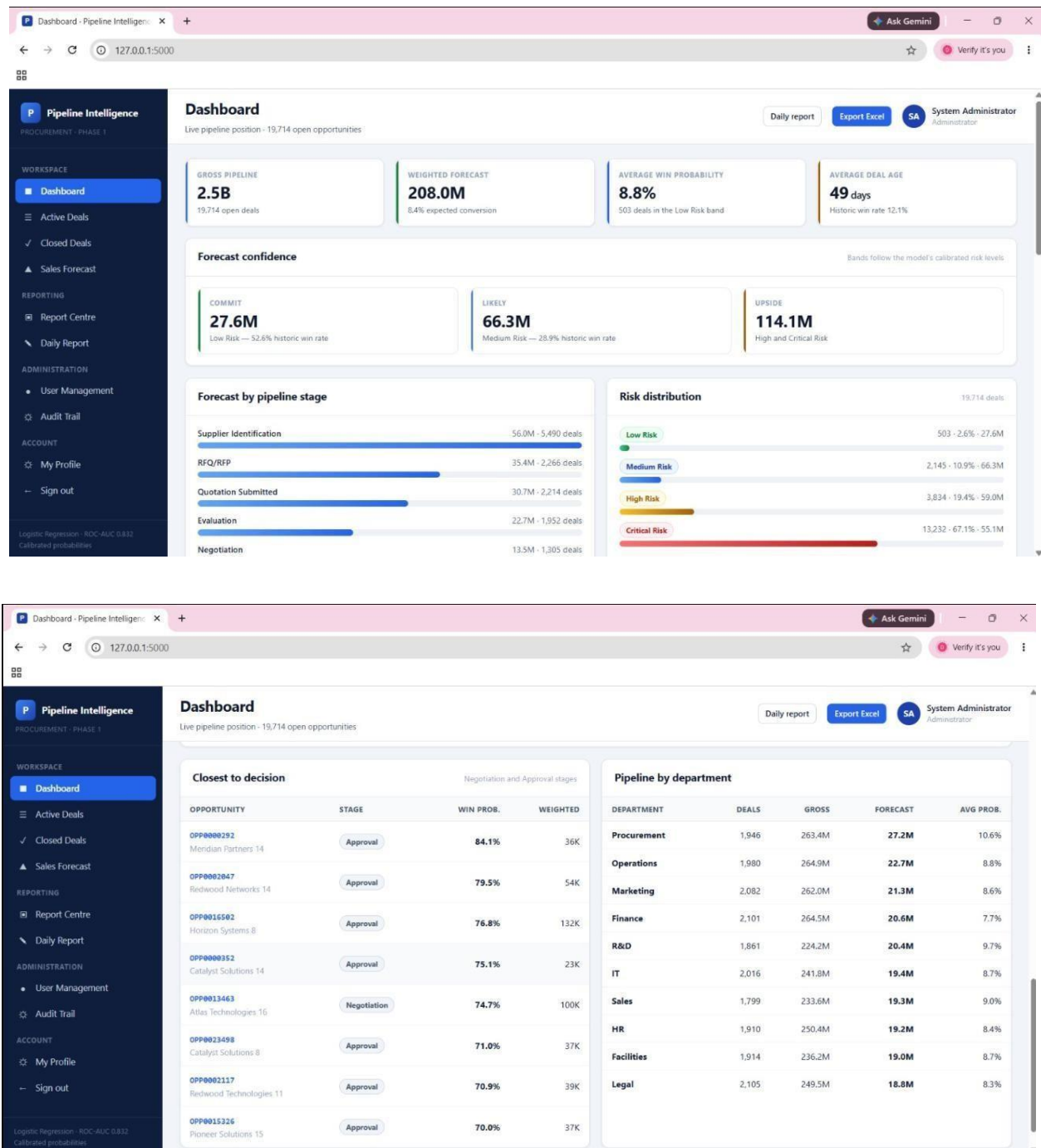


Figure 4.2. Dashboard & Business Intelligence Module

Presents a comprehensive overview of pipeline performance, forecast metrics, risk distribution, and department-wise analytics. Supports executive-level monitoring and decision-making.

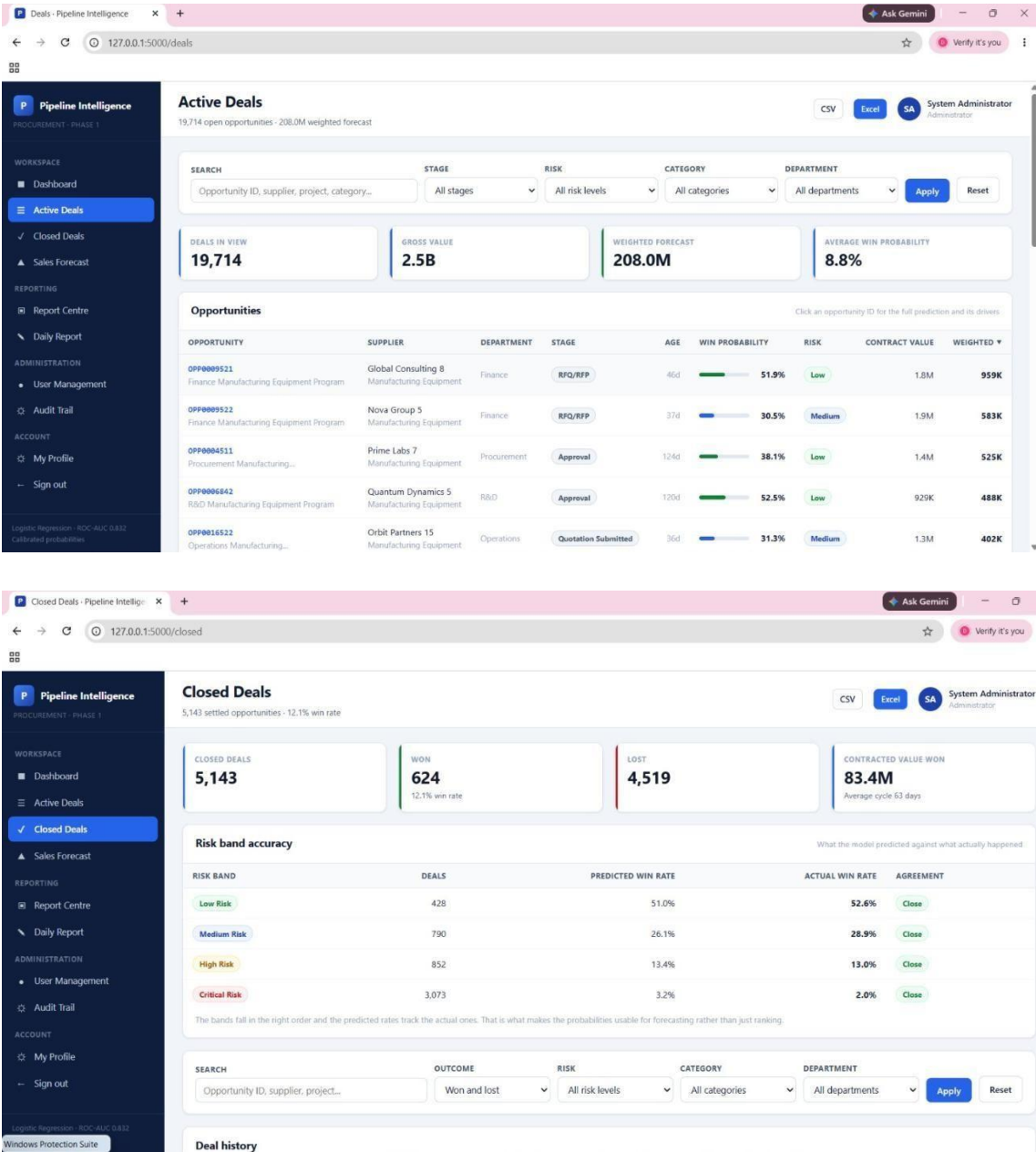


Figure 4.3. Deal Lifecycle Management Module

Manages procurement opportunities from active stages to closure. Tracks deal status, win probability, risk levels, and historical outcomes.

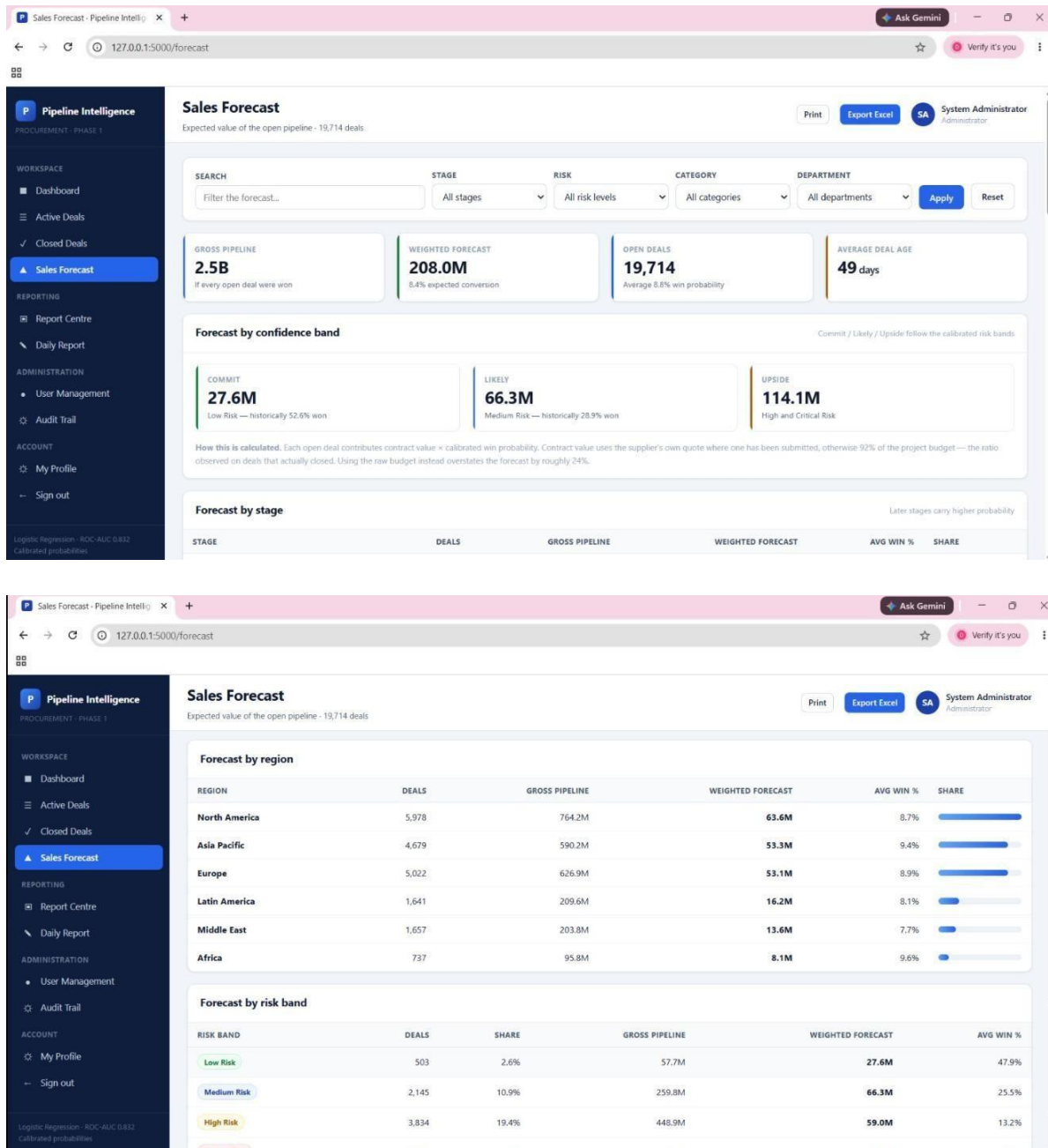


Figure 4.4. Sales Forecasting & Predictive Analytics Module

Uses machine learning-based probability scores to estimate future revenue. Provides forecasts by confidence band, region, stage, and risk category.

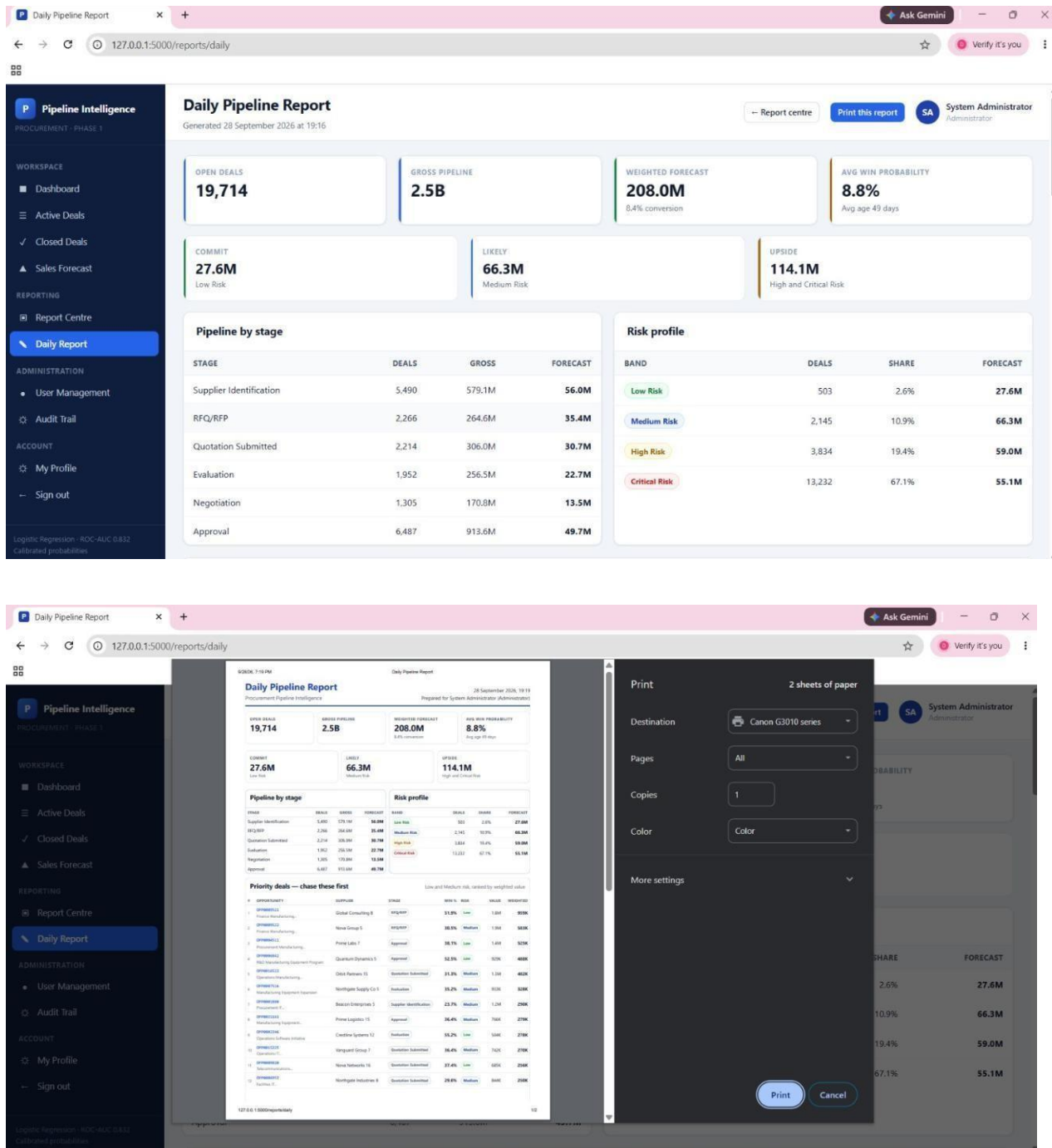


Figure 4.5. Reporting & Document Generation Module

Generates daily pipeline reports and printable summaries for stakeholders. Supports report export, printing, and business performance tracking.

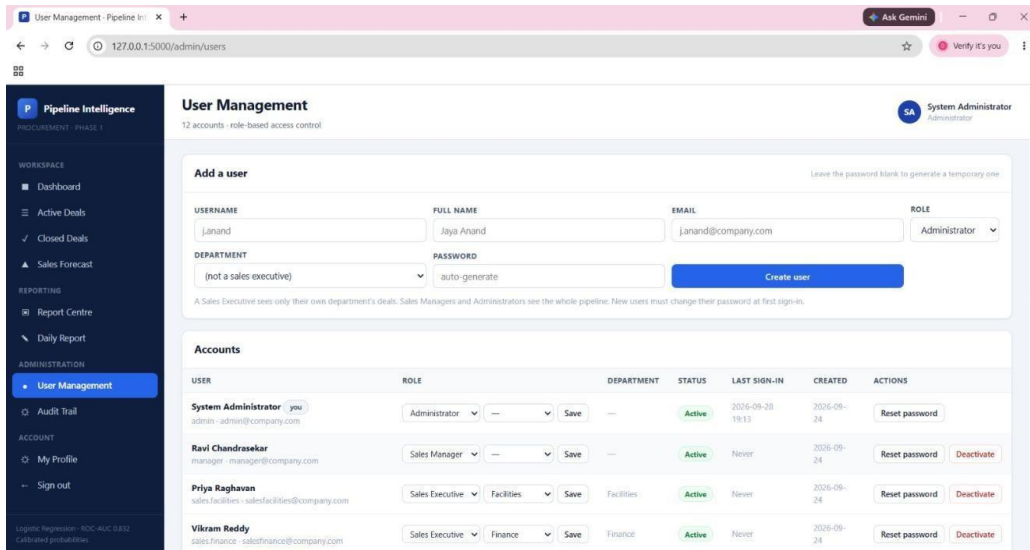


Figure 4.6. User Administration Module

Enables administrators to manage user accounts, roles, departments, and access permissions. Implements centralized user governance.

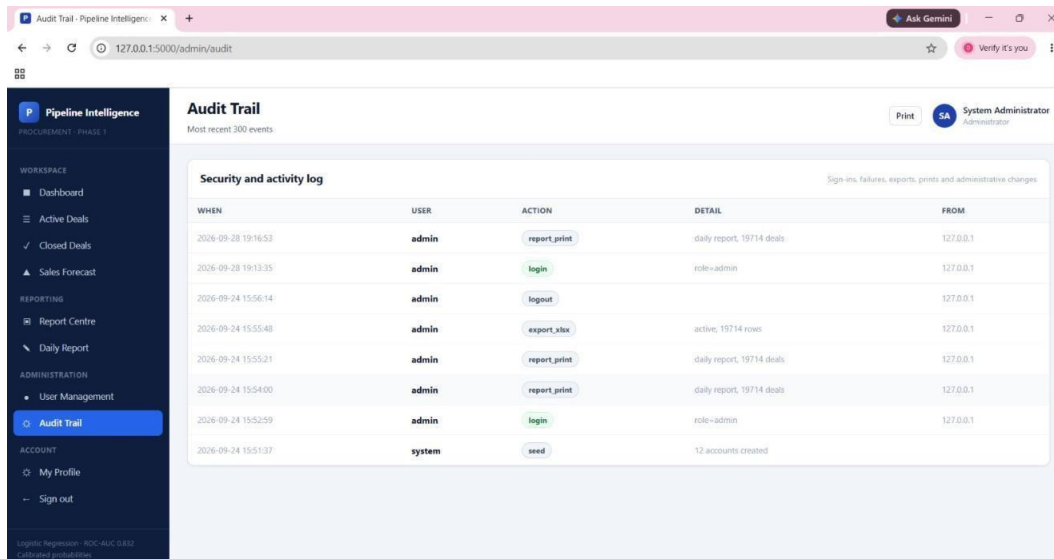


Figure 4.7. Audit & Activity Monitoring Module

Maintains a complete record of system activities, logins, exports, and administrative actions. Enhances security, transparency, and compliance.

CHAPTER 5 IMPLEMENTATION AND RESULT DISCUSSION

This chapter covers about the implementation of the system that has been proposed. It displays the modules that have been built as well as the user interface and the functioning work flow. It also includes the results that are achieved after testing.

5.1 API ENDPOINT SUMMARY

The backend of the system handles all that using a set of REST API end points. These end points are used for authentication, manufacturers, sellers, warranty, ownership transfer, and claims. These API endpoints ensure proper communication between the frontend and the backend.

Method & Path		Description
GET	/api/admin/manufacturers	List all registered manufacturers
POST	/api/admin/manufacturers	Create a new manufacturer entry
GET	/api/admin/manufacturers/{id}	Retrieve details of a specific manufacturer
PUT	/api/admin/manufacturers/{id}	Update manufacturer information
DELETE	/api/admin/manufacturers/{id}	Delete a manufacturer entry

Table 5.1. Admin: Manufacturers register

Method & Path		Description
GET	/api/admin/sellers	List all registered sellers / transactions
GET	/api/admin/sellers/{id}	Retrieve details of a specific seller
POST	/api/admin/sellers	Create a new seller entry
PUT	/api/admin/sellers/{id}	Update seller information
DELETE	/api/admin/sellers/{id}	Delete a seller entry

Table 5.2. Admin: Seller / Transaction register

Method & Path		Description
GET	/api/admin/transfers	List all ownership transfer records
GET	/api/admin/transfers/{id}	Retrieve details of a specific transfer
POST	/api/admin/transfers	Create a new ownership transfer entry
PUT	/api/admin/transfers/{id}	Update transfer information
DELETE	/api/admin/transfers/{id}	Delete a transfer entry

Table 5.3. Ownership Transfers register

Method & Path		Description
POST	/api/admin/sellers/transactions	Add a new seller transaction from dialog
GET	/api/admin/sellers/{id}/products	Retrieve products for the selected seller
GET	/api/admin/sellers/validate-transaction	Validate transaction details (e.g., date, product, price)
GET	/api/admin/sellers/payment-modes	Retrieve available payment mode options

Table 5.4. Admin API endpoints

Method & Path		Description
GET	/api/admin/claims	List all warranty claims across customers
GET	/api/admin/claims/{id}	Retrieve details of a specific claim
POST	/api/admin/claims	Create a new claim entry
PUT	/api/admin/claims/{id}	Update claim status and details
GET	/api/admin/claims/export	Export claims ledger to Excel/CSV

Table 5.5. Global Claims Ledger

5.2 ADMINISTRATIVE DATA MANAGEMENT

The Administrative Data Management module allows administrators to easily manage and update all key procurement and user data for the Sales Orbis platform. Features include management of user profiles, supplier profile management, procurement opening management, general configuration of the system, user access management etc through a single interface. Administrators can create, update and view information and further organize data from the back-end platform thus allowing for smooth control and system administration. Data validation and restrictions have been incorporated to prevent any improper updates to the platform.

5.3 END-TO-END SYSTEM DEMONSTRATION

The End-to-End System Demonstration is a walk through of the Sales Orbis system showing how the module's of the system interact from the initial data load to how the approach assists decision making. The demonstration is a practical overview of how the system uses different algorithms to load procurement data; engineering features; create predictions; assess risk; and generate forecasts. Web reports can be viewed on screen and reports can be written from the Opportunities tab, allowing opportunities to be drilled into, and reports generated. This demonstration demonstrates the interaction of system modules and verifies that system business processes have been successfully completed.

5.4 FUNCTIONAL VALIDATION

The Functional Validation The Functional Validation The Functional Validation is to ensure that the system build meets the functional Requirements and the business objective.

Each module is tested to determine the ability of the platform to undertake the various operations for example user authentication, deal analysis, forecasting, reporting, and administration management. Input and output validation are carried out, including Workflow validation and Role validation. The results obtained confirm that the platform is able to carry out the jobs required on it for example identifying the system handles the role perfectly.

5.5 SYSTEM OUTPUTS AND DECISION SCENARIOS

The Sales Orbis Model Outputs and Decision Scenarios Sales Orbis creates predictive outputs and decision scenarios from these outputs to aid procurement decisions. Output types include probability of winning, risks, value to procurement, forecast overview, and opportunity rankings. These outputs are used to differentiate between opportunities, identify priority opportunities, and identify business risks. Insights produced by dashboards and reports support decision makers to interpret the prediction outputs and prioritize opportunities while aiding planning and allocating resources. This turns predictive insights into actionable insights that can be used by decision makers to aid procurement activities.

5.6 IMPLEMENTATION DISCUSSION AND OBSERVED ISSUES

The Implementation Discussion and Practice Observed Challenges This section describes the real-world challenges we faced during design and implementation of the Sales Orbis platform. These included issues with data quality, dealing with missing data, standardizing features, model calibration, and system integration issues

through various issues were presented during implementation such as the use of pre-processing, and validation methods. Other issues discussed were the issue of monitoring users' access and system performance, and ensuring reliable inter-component communication. These issues notwithstanding, the platform performed according to the expected functionality and performance parameters during implementation.

5.7 LIMITATIONS

The Limitations As with any system, there are current limitations of Sales Orbis that may mean adoption for specific third-party procurement contexts can be further evaluated or customised to fit specific needs. The models have been trained on available prior procurement data and in the event of significant change to business environment, supplier behaviour, or procurement activity these models would need to be retrained. The system makes use of structured data and has not used any additional market intelligence or relevant economic information.

5.8 SUMMARY OF CONTRIBUTIONS

The implemented Phase I contributions can be summarised as follows:

- Developed AI-Matched.com, a SaaS procurement intelligence application predicting deal wins and pipeline analytics.
- Integrated multiple different procurement data sources into a unified leakage-proof analytics engine.
- Deployed feature engineering to extract meaningful procurement and supplier performance indicators.

- Developed and tested machine learning models for forecasting the chance of awarding to supplier for each procurement.
- Produced risk rankings and probability weighted projections.
- Developed and deployed an Explainable AI (XAI) ML model to produce interpretable results and insights for key stakeholders.
- Developed dashboarding, reporting, visualisation products for business intelligence and monitoring applications.
- Implemented data-supported opportunity prioritisation, risk assessments and sourcing planning via a centrally available web-based platform.

CHAPTER 6 CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project phase aimed to build an AI-driven sales pipeline intelligence platform to help predict deal-win probabilities and facilitate data-driven procurement and sales forecasting analysis. We operationalized the entire prediction lifecycle, whereby procurement data is validated, sanitised and made leakage-safe features, opportunities are modelled as snapshots over time, multiple ML models are trained and evaluated, and a best performer is used to generate deal-win probabilities, deal outcome predictions, risk analysis, and expected procurement values. The pipeline's outputs were then visualised in a BI report that is standalone to estimate deal pipeline health, compute open and closed deals, produce individual deal insights, and project future sales pipeline. We answer all the key requirements for dynamic deal-win prediction. Leakage-safe feature engineering is performed based on the information available at each asofdate, and the project-based and temporal data splits make sure that no project overlap is allowed and no future information is used in evaluating the models. Logistic Regression and Decision Tree baselines are tested against more advanced approaches including Random Forest, XGBoost, LightGBM and Cat Boost models. The models are evaluated according to ROC-AUC, PR-AUC, calibrated log-loss, Brier score and temporal robustness. The final model is also inspected in terms of probability calibration and between-training-vs.-test generalisation. It can also give model transparency via SHAP based global and local explanations and diagnosis of prediction behavior at the stage of each pipeline step and each snapshot. Its final prediction module outputs a prediction for each Gold snapshot, and the business report transforms all into actionable sales intelligence, including predicted award rate, risk degree, active-deal details, and forecast value. This bridges the gap between the machine learning pipeline and real-world business

decision support needs, instead of just training the model. The project pointed out various limitations of the existing implementation as well. The data consists of a synthetic dataset with patterns that do not necessarily accurately reflect real procurement behaviour. The labelled outcomes are narrowly sampled over a brief period and pipeline stages such as RFQ/RFP are missing labelled snapshots. The quality of prediction could be reduced by the noise and assumptions in the created data. These limitations point the way for future improvements, such as validation on real data, a broader set of historical pipeline outcomes, additional time-dependent features, better stage coverage, and further model monitoring and calibration.

6.2 FUTURE WORK

The continuous deployment of SALES ORBIS is also building foundation of AI-enabled Sales Pipeline Intelligence and Deal-Win Prediction System. The version is deployed has leakage-proof feature engineering, ML-based award prediction, model scoring, SHAP interpretation, dynamic stage-by-stage prediction, final prediction and a fully 1-independent BI report. The following stages of system improvements will be deployed to have more reliable, interpretable and self-learning prediction system.

- Test the model in practice Validate your models on real data. If you're currently only using a dummy data set, then substitute or add in historical procurement and sales data to test how the trained models handle the reality of actual business. Others Expand the training data for historical deals. More completed opportunities and results at all periods and all stages in the pipeline would support more agile and accurate deal-win prediction outside the current labelled period.
- Improved stage-wise prediction coverage As much as collection of even more labelled data (say RFQ/RFP or other phases that flat user can't do

rather well even with current data) to make it possible for powerful prediction machine to make practical predictions for the entire sales funnel. Development of advanced dynamic prediction models Study of time-aware and sequential approaches that can to some degree track the "journey" of an opportunity across multiple snapshots rather than relying solely on individual snapshots.

- Feature engineering. Add more features (temporal, supplier, competitive, engagement, pricing, pipeline-behaviour) but with lots of leakage controls. Enforce disciplined model monitoring and calibration: Regularly monitor the accuracy of predictions produced by the model, calibration of the model's predicted probabilities, relative changes in the data distribution, and model decay to identify possible model drift and enable a cycle of retraining.
- Create the BI (business intelligence) Dashboard. Expand the existing HTML report with more robust interactive analytics around pipeline value, stage conversion, risk breakout, high/low probability of award, forecast value, supplier performance, and time trends. 2 - Offer automated notifications and tips Include notifications for sizable deals, a shift in the forecast win likelihood, time between activities, and other vital pipeline metrics to direct sales. Improve explainability and support decision-making. - Empower users with additional granular opportunity-level explanations besides the offered SHAP-based explanation, and translate them into. Evaluation of the key model factors provides clear guidance to sales and purchasing teams.
- Automate the whole ML pipeline. Integrate data preprocessing, feature engineering, model training, model evaluation, prediction creation and report creation into a repeatable automated process with only a few clicks to execute the individual python modules. Assess the system regularly Use the evaluation framework above with the new label outcome data to assess and compare the performance across time, pipeline stage and project.

REFERENCES

- [1] J. Yan, C. Zhang, H. Zha, M. Gong, C. Sun, J. Huang, S. M. Chu, and X. Yang, “On Machine Learning towards Predictive Sales Pipeline Analytics,” Proceedings of the AAAI Conference on Artificial Intelligence, vol. 29, no. 1, 2015, doi: 10.1609/aaai.v29i1.9455.
- [2] J. Yan, M. Gong, C. Sun, J. Huang, S. M. Chu, and X. Yang, “Sales Pipeline Win Propensity Prediction: A Regression Approach,” 2015 IEEE International Conference on Information and Automation, 2015, doi: 10.1109/INM.2015.7140393.
- [3] V. Eitle and P. Buxmann, “Business Analytics for Sales Pipeline Management in the Software Industry: A Machine Learning Perspective,” Proceedings of the 52nd Hawaii International Conference on System Sciences (HICSS), 2019.
- [4] M. Lambert, “Sales Forecasting: Machine Learning Solution to B2B Sales Opportunity Win-Propensity Computation,” Master’s Thesis, National College of Ireland, 2018.
- [5] A. Rezazadeh, “A Generalized Flow for B2B Sales Predictive Modeling: An Azure Machine-Learning Approach,” Forecasting, vol. 2, no. 3, pp. 1–17, 2020, doi: 10.3390/forecast2030015.
- [6] S. Mortensen, M. Christison, B. Li, A. Zhu, and R. Venkatesan, “Predicting and Defining B2B Sales Success with Machine Learning,” 2019 Systems and Information Engineering Design Symposium (SIEDS), 2019, doi: 10.1109/SIEDS.2019.8735638.
- [7] T. Woźniakowski, “Machine Learning Based Predictions of Sales Leads: Proof of Quality from Polish Business-to-Business Company,” Methods of

Quantitative Analysis in Economics, vol. 26, no. 2, pp. 70–84, 2025, doi: 10.22630/MIBE.2025.26.2.7.

- [8] H. Ahaggach, L. Abrouk, and E. Lebon, “Systematic Mapping Study of Sales Forecasting: Methods, Trends, and Future Directions,” *Forecasting*, vol. 6, no. 3, pp. 502–532, 2024, doi: 10.3390/forecast6030028.
- [9] M. L. Dominic, P. S. Venkateswaran, L. Thamma Reddi, S. Rangineni, R. Regin, and S. Suman Rajest, “The Synergy of Management Information Systems and Predictive Analytics for Marketing,” in *Data-Driven Decision Making for Long-Term Business Success*, IGI Global, pp. 49–63, 2024, doi: 10.4018/979-8-3693-2193-5.ch004.
- [10] M. S. Kasem, M. Hamada, and I. Taj-Eddin, “Customer Profiling, Segmentation, and Sales Prediction Using AI in Direct Marketing,” *Neural Computing and Applications*, vol. 36, pp. 4995–5005, 2024, doi: 10.1007/s00521-023-09339-6.
- [11] Y. Nuthakki, “Smart Opportunity Scoring in Salesforce Sales Cloud: A Machine Learning Approach,” *Sarcouncil Journal of Multidisciplinary*, vol. 5, no. 7, pp. 662–668, 2025.
- [12] P. Raja, “Next-Gen Sales Forecasting in CRM Through AI and Pipeline Intelligence in Dynamics 365,” 2025.
- [13] S. Henna, S. K. Kalliadan, and M. Amjath, “Optimizing B2B Customer Relationship Management and Sales Forecasting with Spectral Graph Convolutional Networks: A Quantitative Approach,” *Quantitative Finance and Economics*, vol. 9, no. 2, pp. 449–478, 2025, doi: 10.3934/QFE.2025015.
- [14] K. Sheahan, “Developing and Empirically Testing a Sales Pipeline Execution Process Framework,” Ph.D. Thesis, Technological University Dublin, 2020.